

Structure of single and mixed alkali Li–Rb borate glasses by neutron diffraction

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Abstract

We have compared the local structure of Li–, Rb– and Li–Rb diborate glasses using neutron diffraction and reverse Monte Carlo simulations. The fraction of tetrahedral boron decreases as the Rb content increases, but is similar in the single and mixed alkali glasses at diborate compositions. Rb borate glasses exhibit an extensive cation ordering with Rb–Rb correlations at 4.7 and 7 Å. The environment of alkalis is slightly modified between the single and mixed glasses while their relative distribution is essentially random. This indicates that the alkali environment is affected by the presence of a different alkali and this cation–cation interaction could be a fundamental mechanism, together with the site mismatch, to explain the mixed alkali effect.

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1. Introduction

Properties related to ionic transport drastically drop when a fraction of the mobile ions is substituted by another type of mobile ions, giving rise to the so-called mixed alkali effect (MAE). [1,2]. An example of this effect is the deep minimum of the electrical dc conductivity when one type of alkali ion is progressively replaced by another, at constant overall cation content [3]. Many models have been proposed to explain MAE but a complete comprehensive view of this phenomenon is still lacking. The dynamic structure model (DSM) predicts that ions preferentially migrate via pathways formed by ion-specific sites [4–6]. Therefore, some conduction pathways are blocked in the mixed alkali glasses due to mismatch of different cation sites. This model is based on the assumption that the local

environment of alkalis is unaffected by the presence of a second alkali. However, recent NMR data suggest that it may not be the case: in mixed alkali glasses, smaller and larger cations suffer site compression and expansion, respectively [7]. It was also proposed that MAE can be due to network modifications induced by alkali ions of different size and an energy distribution of sites with a preferential occupancy of the low energy sites by the smaller cations [8]. Some structural aspects are critical to improve the understanding of cation transport in glasses: local oxygen environment and spatial distribution of the mobile alkali metal ion, relative cation mixing in ternary glasses, influence of the network structure on cation mobility.

MAE has been recently widely studied in alkali borate glasses [9–16], for which the motion of alkali ions is appreciable already far below T_g . The dc conductivity of a Na-borate glass is larger by more than one order of magnitude than that of the corresponding Rb-borate glass, indicating that the smaller Na ions move faster than the larger Rb ions [16]. The diffusion of ions and the electrical

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conductivity operate through successive jumps from and toward the sites available within the essentially static borate network. Therefore the network structure may play a major control on long-range transport.

In pure B_2O_3 , boron atoms are localized in BO_3 units forming a continuous random network (CRN). The incorporation of M_2O alkali oxide gives rise to BO_4 tetrahedral units, where the negative charge of the tetrahedron is charge-compensated by alkali ions. As the concentration of non-bridging oxygen atoms (NBO's) is negligible up to about 30 mol% alkali oxide content, the addition of alkali oxide to B_2O_3 improves the stability of the CRN by cross-linking BO_3 and BO_4 units [17]. Beyond this concentration, alkalis begin to depolymerize the network with the formation of a significant amount of NBO's.

In the present contribution, we present new data on the structure of single- and mixed alkali Li–Rb borate glasses, using neutron diffraction and Reverse Monte Carlo simulations. Li–Rb borate glasses were chosen because of the importance of MAE and the large difference in the ionic radius between the two alkalis. Single and mixed alkali diborate glasses exhibit a similar fraction of tetrahedral boron, but with a slightly modified environment of alkalis. As the alkali environment is affected by the presence of a different alkali, this cation–cation interaction could be a fundamental mechanism at the origin of the MAE.

2. Experimental

2.1. Glass preparation

Glasses were synthesized under dry conditions, using isotopically enriched $^{11}B_2O_3$ to avoid the high neutron absorption of ^{10}B nuclei, and reagent grade powders of 7LiCO_3 (99.94% 7Li) and $RbCO_3$, according to the compositions in Table 1. The starting mixtures were decarbonated at 700 °C during 12 h, then melted in a covered Pt crucible at 1020 °C for 20 min. The melts were quenched by rapid immersion of the bottom of the crucible in water. This procedure was repeated twice to ensure good homogeneity. A previous chemical analysis of Li-diborate glasses prepared under the same conditions indicated only a weak deviation from the nominal composition [18]. Therefore, we used the nominal composition in the data analysis. Densities, measured by an Archimedes method with toluene as liquid ref-

erence, are reported in Table 1 and are in good agreement with those reported in the literature [16,19,20]. The samples were stored in a desiccator prior to experiments to avoid water absorption.

2.2. Neutron diffraction experiments and data analysis

The neutron diffraction experiments were performed on the 7C2 diffractometer (Orphée reactor, Laboratoire Léon Brillouin, France). This instrument uses hot neutrons of wavelength 0.697 Å, giving access to a Q -range of 0.5–16 Å^{−1}. Glasses were powdered just before the measurement in order to avoid hydration, and set in a cylindrical vanadium cell, at the center of a cylindrical vanadium furnace. The data were treated with standard procedures [21]. The inelastic scattering contribution in the structure factors was corrected using the MCGR code, which removes a quadratic background from the data [22].

The total structure factor $F(Q)$ is the distinct scattering term of the total neutron cross section and can be written in the Faber–Ziman formalism as

$$F_T(Q) = \sum_{\alpha,\beta}^{n,n} c_\alpha c_\beta b_\alpha b_\beta (A_{\alpha\beta}(Q) - 1), \quad (1)$$

where $A_{\alpha\beta}$ are the Faber–Ziman partial structure factors.

The Fourier transformation of $F(Q)$ gives the total radial distribution function $G(r)$, which is the neutron weighted sum of all the partial pair distribution functions $g_{\alpha\beta}(r)$.

$$G(r) = \sum_{\alpha,\beta}^{n,n} c_\alpha c_\beta b_\alpha b_\beta (g_{\alpha\beta}(r) - 1). \quad (2)$$

We define the total correlation function $T(r)$ as

$$T(r) = 4\pi\rho_0 r \left[G(r) + \left(\sum_i c_i b_i \right)^2 \right]. \quad (3)$$

2.3. RMC simulations

Reverse Monte Carlo (RMC) simulations have been carried out on the single Li and Rb diborate glasses and on the mixed Li–Rb glass using the neutron diffraction data. The RMC method consists in fitting the neutron diffraction data to obtain three-dimensional models that agree quantitatively with the experimental data (provided that the data do not contain significant systematic errors) and the physical constraints applied [23]. Starting with a hard-sphere Monte Carlo (HSMC) simulation results in a more disordered model than starting with molecular dynamics models [24]. However, in the present case, the advantage of a HSMC simulation is to be able to control the boron coordination. In the RMC simulations, 60% and 40% B atoms were constrained to be 3- and 4-coordinated. In this way, the fraction of BO_4 units corresponds to that estimated from a Gaussian fit of neutron diffraction data (see below)

Table 1
Glass composition, density and glass transition temperature of the glasses

Composition	B_2O_3 (mol%)	Li_2O (mol%)	Rb_2O (mol%)	d (g cm ^{−3})	T_g (K)
7LB2	66.66	33.34		2.23	780 ^a
7LRbB2	66.66	16.67	16.67	2.64	697 ^a
RbB2	66.66		33.34	2.92	707 ^{a,b}
RbB3	75.00		25.00	2.74	720 ^b
RbB4	80.00		20.00	2.54	707 ^b

^a Gao (2006).

^b Berkemeier et al. (2005).

and from NMR data [15]. No coordination constraint was imposed on the alkali ions. Closest approach distances were fixed from the ionic radii. The size of the simulation box, containing 3900 atoms, was set to satisfy the macroscopic density of the system.

3. Results

3.1. Total structure factors

The total structure factors of the glasses are presented in Fig. 1. At large Q values ($Q > 6 \text{ \AA}^{-1}$), the signal is dominated by correlations within planar BO_3 triangles and BO_4 tetrahedra. The features at low- Q values are mainly related to the intermediate range order (IRO), which may involve preferred arrangements between borate units and/or alkali ions. The low- Q range is characterized by a peak at $1.53 \text{ \AA}^{-1}$ for the 7LB2 glass. In the mixed Li–Rb glass, two contributions emerge at $1.25 \text{ \AA}^{-1}$ and $1.88 \text{ \AA}^{-1}$. In the pure Rb glasses, the first peak is shifted at $1.18 \text{ \AA}^{-1}$ for RbB2 and $1.15 \text{ \AA}^{-1}$ for RbB3 and RbB4. The intensity of this peak decreases in the mixed glass compared to the peak at $1.53 \text{ \AA}^{-1}$ in 7LB2 and then increases and becomes sharper in the Rb glasses. The second contribution is at $2 \text{ \AA}^{-1}$ in RbB2 and shifts gradually to $1.87 \text{ \AA}^{-1}$ in RbB3 and $1.80 \text{ \AA}^{-1}$ in RbB4. A further, weak peak occurs near $3.3 \text{ \AA}^{-1}$ in 7LB2; it becomes broader and shifts to lower Q values as Rb is added. These changes are directly related to the presence of Rb and indicate that addition of Rb_2O modifies the IRO. A linear combination of RbB2 and 7LB2 $S(Q)$ functions reproduces qualitatively the structure factor of the mixed glass (dashed curve in Fig. 1).

3.2. Total correlation functions

The total correlation functions of the 7LB2 glass are dominated by the contributions of the borate network

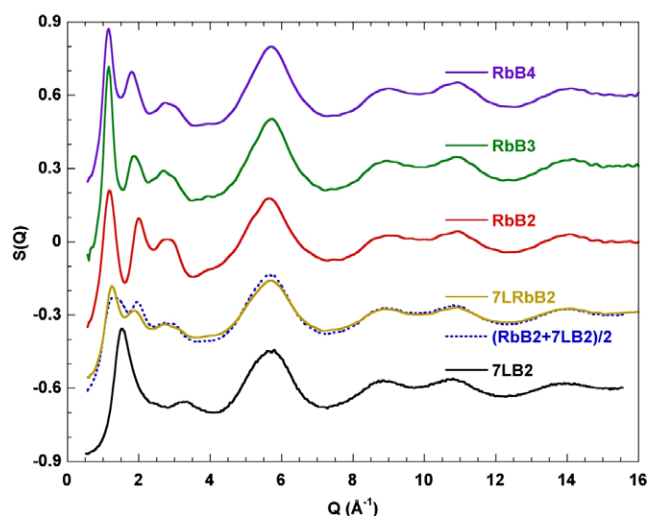


Fig. 1. Total structure factors of the single and mixed alkali glasses. The curves have been displaced for clarity. The dashed line is the average of the $S(Q)$ for RbB2 and 7LB2.

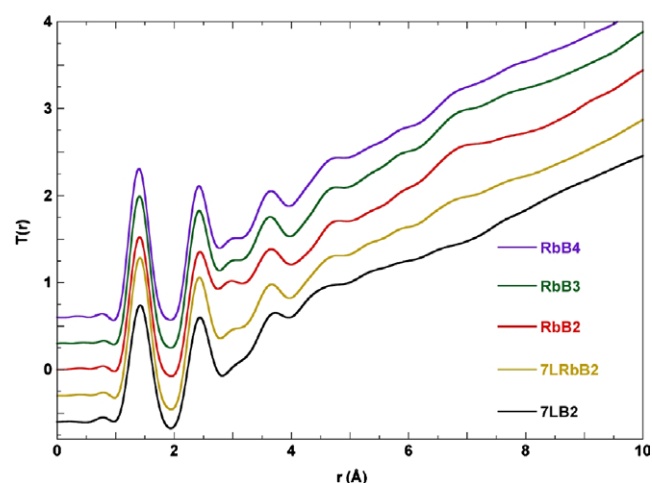


Fig. 2. Total correlation functions of the single and mixed alkali glasses. The curves have been displaced for clarity.

(Fig. 2). Structural contributions are clearly discernible below $6 \text{ \AA}$. The two prominent peaks at $\sim 1.40 \text{ \AA}$ and at $\sim 2.45 \text{ \AA}$ are attributed to the B–O and O–O correlations involving BO_3 and BO_4 units, respectively, with a minor contribution from the first B–B correlation occurring within the second peak [18]. The well-defined third peak at $3.65 \text{ \AA}$, and the fourth peak at $4.5 \text{ \AA}$ are assigned to B–O2, where O2 refers to the second O neighbors, and O–O2, respectively. They indicate that network ordering exists beyond the basic structural units, in the presence of superstructural units (borate rings) which are well known in borate glasses. With the addition of Rb ions, a contribution appears at $2.98 \text{ \AA}$ and becomes increasingly intense as the Rb content increases. This peak can be attributed to Rb–O correlations. Contributions are also apparent in Rb glasses around 4.7 and $7 \text{ \AA}$ and, by comparison with former investigation of heavy elements in borate glasses, they can be ascribed to cation–cation distances [25].

Quantitative information on the coordination of boron atoms are obtained by Gaussian fitting of the first B–O peak. The fitting parameters are the mean B–O distance (R), the Debye–Waller factor (σ), and the coordination number (N). Best fits were obtained using two Gaussian functions modeling boron in triangular (N_1) and tetrahedral sites (N_2) (Table 2). In the fitting procedure, these two values are linked by the relation $\frac{N_1}{3} + \frac{N_2}{4} = 1$. Errors on the fitting procedure have been explained previously [18].

3.3. RMC simulations

The neutron structure factors calculated using the RMC models are compared with the experimental data in Fig. 3. The sum of the RMC partial structure factors due to the network (B–B, B–O and O–O pairs) and due to the alkalis (M–M, M–O, M–B) are also presented in Fig. 3 to help the interpretation of the low- Q features. The peaks at $1.54 \text{ \AA}^{-1}$ for LB2 and at $1.17 \text{ \AA}^{-1}$ for RbB2 are associated with the

Table 2
Distances (*R*), coordination numbers (*N*) and Debye–Waller factors obtained by Gaussian fitting or extracted from the RMC simulations

	<i>R</i> ₁ (Å) ±0.005 Å	<i>N</i> ₁ ±0.02	<i>σ</i> ₁ (Å) ±0.002 Å	<i>R</i> ₂ (Å) ±0.005 Å	<i>N</i> ₂ ±0.02	<i>σ</i> ₂ (Å) ±0.002 Å	<i>N</i> ₄ ±0.02 (neutron)	<i>N</i> ₄ (NMR)
7LB2	1.380	1.62	0.06	1.470	1.84	0.096	0.44	0.41 ^a
7LRbB2	1.365	1.75	0.035	1.477	1.67	0.045	0.42	
RbB2	1.360	1.71	0.055	1.475	1.72	0.065	0.43	0.43 ^b
RbB3	1.362	1.85	0.044	1.472	1.53	0.06	0.38	
RbB4	1.362	2.1	0.05	1.477	1.2	0.065	0.30	0.30 ^b

^a Ratai et al. (2002).
^b Epping et al. (2005).

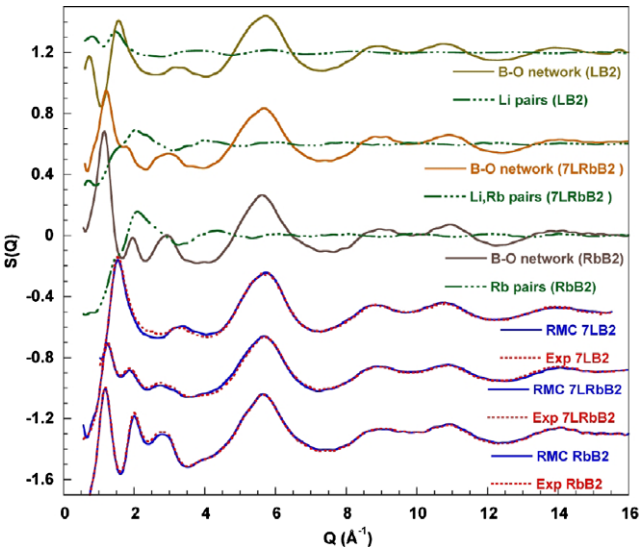


Fig. 3. Comparison of the structure factor obtained by neutron diffraction and RMC simulations for the lower curves and sum of the partial structure factors due to the network (B–B, B–O and O–O pairs) and due to the alkalis (M–M, M–O, M–B) for the 7LB2, 7LRbB2 and RbB2 glasses.

pairs involving the borate network (B–B, B–O and O–O pairs). In the case of LB2, a contribution of Li-pairs is also noticeable in the peak at $1.54 \text{ \AA}^{-1}$, which is mainly due to the Li–B pair and gives rise to a strong dependence upon the Li scattering length, when comparing Li isotopic substitution neutron diffraction data [26]. This points out the structural relationship between Li ordering and borate network topology, based on an alternate ordering of Li and B/O atoms, whereas a weaker association exists between the borate network and the Rb ions. The peaks between $2\text{--}4 \text{ \AA}^{-1}$ are related to an interplay between the pairs associated with the B–O network and those associated with the alkali ions.

In the RMC simulation of LB2, we used neutron scattering data from 7LB2 and 6LB2 glasses that contain ^7Li and ^6Li , respectively. Giving a cut-off distance of $2.5 \text{ \AA}$, Li atoms are surrounded in average by 4.2 oxygen atoms. In the mixed glass, the coordination number for Li decreases to 3.1 oxygen neighbors (but in the case of the 7LRbB2 glass, the Li–O partial is weak and thus the coordination number is less constrained than in 7LB2). If we consider Rb atoms with a cut-off distance of $3.5 \text{ \AA}$, we also observe

a slight change in the coordination number from 5.4 in RbB2 to 7.1 in LRbB2. These partials are not all equivalently constrained in the simulation due to their different neutron weights and thus uncertainties in the coordination number exist.

4. Discussion

The structures at low-*Q* values correspond to a medium-range ordering, whose microscopic origin is not well understood. RMC simulations indicate a strong contribution from the borate network. Since the position of these peaks in *S(Q)* corresponds in the real space to inversely proportional characteristic distances, the peak at low-*Q* for RbB2 indicates larger distances in real space than in LB2. This is thus a direct evidence for the network expansion as larger Rb ions are substituted to smaller Li ions. The mixed Li–Rb glass presents an intermediate case.

All the features present in the mixed Li–Rb glass below $4 \text{ \AA}^{-1}$ are obtained by averaging the structure factors of 7LB2 and RbB2 (Fig. 1), though the peak positions and intensities are not exactly reproduced. This suggests that the topology of the Li–Rb network is mainly determined by the network topologies of the single glasses. However, that some discrepancies exist in the position and broadening of the peaks indicates that mixed alkali glasses do not simply result from an average mixing of single alkali glasses. Indeed, the densities of the mixed alkali glass are closer to that of the Rb single glasses [19,20].

Our diffraction data give interatomic distances, which are in good agreement with those previously obtained for BO_3 and BO_4 units (Table 2). The proportion of *N*₁ increases as the Rb content decreases and conversely the proportion of *N*₂ decreases. *N*₁ and *N*₂ values are in good agreement with those determined from NMR measurements [15]. There is no indication that the mixed alkali glass presents a different amount of BO_4 tetrahedra than single alkali glasses. These results agree with ^{11}B MAS NMR experiments for which the fraction of BO_4 (*N*₄) in Na–Rb glasses is primarily influenced by the total cation content and that mixing Na and Rb ions does not modify *N*₄ [7,15].

The Li environment has been previously investigated in LB2 glass, showing that the mean Li–O distance is

$1.99 \pm 0.01 \text{ \AA}$, related to an average distorted tetrahedral site [26]. The Gaussian fit of the Li–O peak gives a large value for the coordination number ($N = 4.9$) and for the Debye Waller factor ($\sigma^2 = 0.090 \text{ \AA}^2$), which are affected by important uncertainties due to systematic errors. Molecular Dynamics (MD) simulations on Li borate glasses indicate that Li cations are located in various distorted sites with 4 to 7–8 oxygen neighbors [27]. RMC simulation indicates that the Li coordination number decreases from 4.2 for LB2 to 3.1 for 7LRbB2 glass, while Rb coordination number increases from 5.4 in RbB2 to 7.1 in LRbB2. There is thus a clear trend that reproduces the recent observation by NMR heteronuclear correlation investigation of Na–Li borate glasses [7]: an expansion of the sites of the large alkalis and a compression of the sites of the small alkalis. Therefore, the alkali environments can be altered when they are mixed, an indication of mutual interactions between dissimilar alkali ions. The change in the ion environment, though rather small, should be more important as the size difference between alkalis increases. It should also have implication in the borate network topology.

Contrary to silicates, the spatial cation distribution is random and clustering effects are absent in alkali borate glasses [7,15,28]. This is also verified in mixed alkali borate glasses but preferences for like-cation interactions have been evidenced and present a maximum near $\text{Na}_2\text{O}/\text{Rb}_2\text{O} = 0.4$, indicating a small deviation from a purely statistical cation mixing at such compositions [15]. A more homogeneous cation mixing is observed with smaller alkali cations [7]. In the RMC simulations, there is no general trend on the alkali–alkali partial pair correlation functions, indicating an homogeneous distribution of alkali ions in the borate network. Indeed, alkali ions are associated with BO_4 tetrahedra for which direct BO_4 – BO_4 connections are partially avoided [29]. This avoidance prevents BO_4 tetrahedra clustering, and consequently alkali clustering. Moreover, we do not observe any preferential tendency for pairing of similar ions and close Li–Rb correlation exists.

The alkali–alkali interactions, observed in the above results, support recent findings that alkali mobility is affected by the close proximity of another type of alkali [7,30]. Such alkali–alkali interactions could be responsible to the mixed alkali effect, together with the energy site mismatch. Considering this latter mechanism, the difference in local structural environment for two types of alkali ions is enhanced in the mixed glass, which increases the energy mismatch for ion jumping between dissimilar alkali sites.

5. Summary

The structures of single Li and Rb and mixed Li–Rb glasses have been investigated by neutron diffraction and RMC simulations. The boron coordination has been evaluated from the first B–O peak in the diffraction data and shows good agreement with NMR results. The fraction of tetrahedral boron, N_4 , decreases in Rb glasses as the

Rb content increases and is not modified in the mixed glass compared to the single alkali glasses. The Rb–O distances is at $\sim 2.9 \text{ \AA}$ and correspond to larger site (7–5 oxygen neighbors) than the Li site (4–3 oxygen neighbors). In RMC simulations, a trend is observed for a compression of the Li sites and an expansion of the Rb sites in the mixed glass. The introduction of a different alkali in the glass network can thus modify the alkali environment. This interaction of the two type of alkali can be significant to understand the mixed alkali effect. More details on the alkali environment and alkali distribution will be obtained by isotopic substitution of Li and structural changes with temperature will be examined since the ionic diffusion is thermally activated.

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